Extreme Barrier method

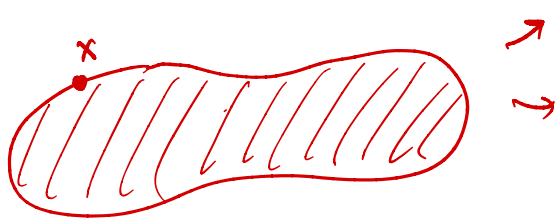
$$\min f(x) \quad \text{s.t. } x \in \Omega \subset \mathbb{R}^n \quad \longrightarrow \quad \min \bar{f}(x) \quad \text{s.t. } x \in \mathbb{R}^n$$

$$\bar{f}(x) \equiv \begin{cases} f(x) & \text{if } x \in \Omega \\ \infty & \text{if } x \notin \Omega \end{cases}$$

$\bar{f} : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$

extended-value function

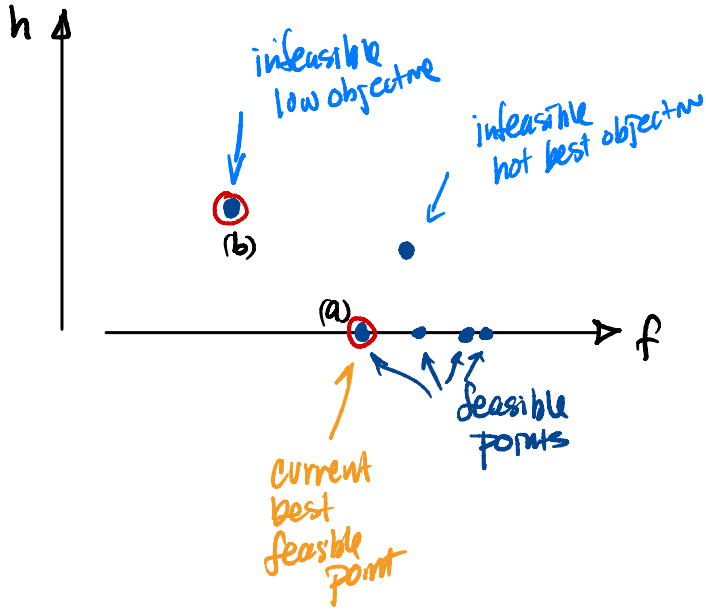
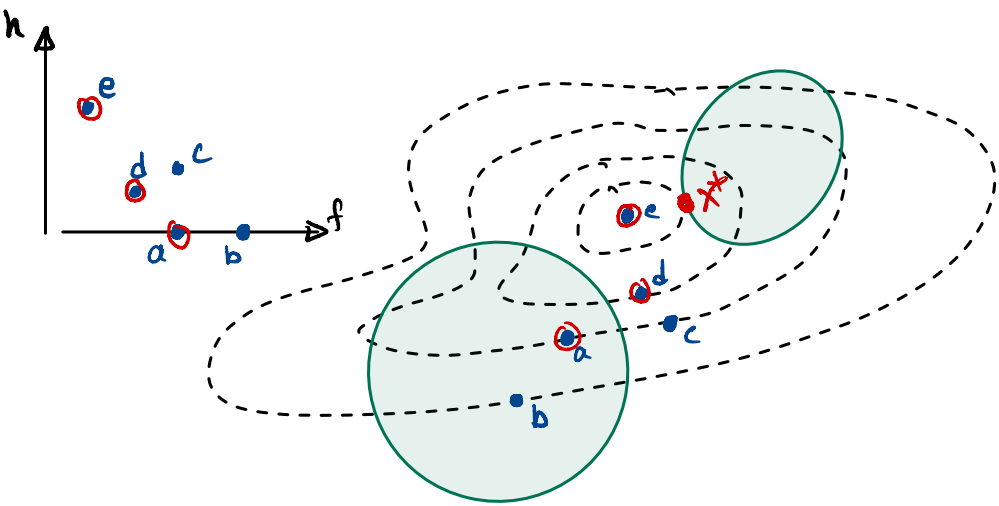
- $\bar{f}$ inherits properties of f on $\operatorname{int}(\Omega)$.
- If $f(x)$ is convex on Ω then $\bar{f}(x)$ is convex on $\mathbb{R}^n$.
- Useful in DFO application with controlled step size and level set boundedness assumptions.



- Tom often applies this technique.

Filter Methods

Treat the problem as a two-objective problem. Every evaluated point can be referenced to its objective value (f) and constraint violation (h).



Concept: search around points (a) and (b)